Detection of Saccades and Quick-Phases in Eye Movement Recordings with Nystagmus

Sai Akanksha Punuganti
Department of Biomedical Engineering
Johns Hopkins University
Baltimore, MD, USA
spunuga1@jhu.edu

Jorge Otero-Millan
Department of Neurology
Johns Hopkins University
Baltimore, MD, USA
jotero@jhu.edu

ABSTRACT

Benign Paroxysmal Positional Vertigo (BPPV) is the most common cause of vertigo and dizziness. Patients with those symptoms can be diagnosed by the presence of a specific pattern of nystagmus during the Dix-Hallpike maneuver. However, almost half of dizzy patients visiting Emergency Department (ED) are misdiagnosed, leading to significant morbidity and high medical costs. This can be attributed to the lack of specialized expertise of front-line physicians and to the lack of validated automatic commercial devices and software for nystagmus detection and quantification. Here we aim to enhance saccade detection thereby improving automatic nystagmus quantification. The proposed method is evaluated on a nystagmus dataset recorded from patients in the ED as they undergo the Dix-Hallpike maneuver. Additionally, the proposed method is also tested on a publicly available saccade dataset and compared with state-of-the-art eye movement detection methods.

CCS CONCEPTS

• **Theory of computation** → **Design and analysis of algorithms**; • **Computing methodologies** → *Unsupervised learning*; • **Applied computing** → *Health care information systems.*

KEYWORDS

BPPV, Nystagmus, Quick-Phases, Saccade Detection, Vertigo

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1 INTRODUCTION

About 3.3% of the chief complaints presenting in Emergency Departments (ED) are concerned with dizziness and vertigo [Newman-Toker et al. 2008]. Among these, the largest proportion corresponds to patients affected with benign peripheral disease of the inner ear, while some patients can suffer from central brain disorders such as stroke. Misdiagnosis can result in unnecessary testing and costs when patients with peripheral disease are thought to have a central cause. Further, it can also cause harm to patients with missed strokes. Almost half of the dizzy patients visiting the ED are misdiagnosed, leading to significant morbidity and financial burden to the patients [von Brevern et al. 2006]. Eye movements test such as HINTS (Head Impulse test, Nystagmus, and Test of Skew) are key tools to avoid such diagnostic errors [Kattah Jorge C. et al. 2009]. Because most ED practitioners are not trained to perform or interpret these tests, there is a need for automatic quantitative analysis of eye movements in patients with dizziness and vertigo.

Nystagmus is a pattern of involuntary eye movements alternating slow eye drifts in a constant direction and quick-phases (QPs, which are similar to saccades) where the eye jumps in the opposite direction (Figure 1). Causes of nystagmus include central neurological disorders, visual deficits, and peripheral disorders [Tusa 1999]. About 17% to 42% of patients with vertigo are given a final diagnosis of BPPV [Hanley et al. 2001; Katsarkas 1999] and it is more often seen in patients above 60 years old [Lempert and Neuhauser 2009]. BPPV is caused by the presence of debris (otoconia) within the semicircular canals and is characterized by a brisk nystagmus lasting around 30s induced by the Dix-Hallpike Maneuver [Dix and Hallpike 1952], in which experts orient patients' head into prescribed positions. Here we will focus on automatic quantification of nystagmus recorded at the bedside during the Dix-Hallpike Maneuver with the objective of identifying patients suffering with BPPV. Specifically, we will improve the methods that identify quick-phases of nystagmus.

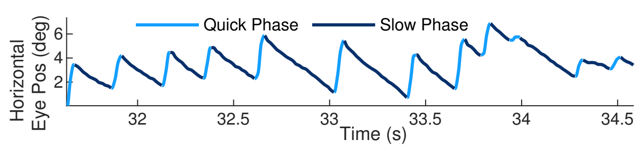

Figure 1: Example of nystagmus trace.

2 RELATED WORK

Given the similarities between quick-phases and saccades, we can rely on saccade detection algorithms to detect quick-phases. These methods can fall into four classes: threshold, probability, statistical, and machine learning based algorithms. Threshold based algorithms include Identification by Velocity Threshold (IVT) [Erkelens and Sloot 1995], Identification by Dispersion Threshold (IDT), EK [Engbert and Kliegl 2003], I-VVT [Komogortsev and Karpov 2013], and NH [Nyström and Holmqvist 2010]. These algorithms incorporate

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thresholds determined by either physiology, experimental conditions or adaptively from the data. Recently published probabilistic approaches [Daye and Optican 2014; Mihali et al. 2017; Santini et al. 2015; Toivanen et al. 2015] make use of Bayesian estimators and generative models of eye movements. Statistical based algorithms [Korda et al. 2018; Mould et al. 2012; Mulligan 2018; Veneri et al. 2011] employ thresholds on statistical tests within windows of the signal. Recent years have also seen Machine Learning based algorithms being applied on eye tracking data [Bellet et al. 2018; Hessels et al. 2017; Hoppe and Bulling 2016; Juhola et al. 2013; Otero-Millan et al. 2014; Pander et al. 2014; Sheynikhovich et al. 2018; Startsev et al. 2019].

Most of the above methods have not been tested in data with nystagmus or recorded at the bedside. Here, we are interested in analyzing eye movement recordings obtained at bedside which pose additional challenges and may be corrupted with patient head movements and significant noise due to other sources (environmental, goggle slippage, systematic etc.,) at a higher rate than controlled recordings performed in a laboratory. Also, physiological parameters for eye movements in presence of nystagmus can differ considerably from that of normal recordings such as an absence of steady fixations, a higher rate of saccade (quick-phases) or a biased distribution of saccade (quick-phase) directions. Hence, the present study explores development of QP detection method in presence of nystagmus.

3 DETECTION OF QUICK-PHASES (QP)

This method was designed with three main principles. First, it should perform careful pre-processing to avoid false negatives in poor quality bedside recordings from patients. Second, it should be accurate in the presence of nystagmus, when eye velocity in between quick-phases can be as high as the velocity of small saccades. Third, detection should be unsupervised with design parameters based on physiology of the ocular motor system. For pre-processing we identify any portion of data with dynamics outside physiological ranges and remove them. To account for nystagmus, we calculate an approximation of the slow-phase velocity and subtract it from the instantaneous eye velocity so that the only periods that deviate significantly from zero are the quick-phases or saccades (Figure 2). Finally, for unsupervised clustering we follow a similar strategy for peak (saccade candidate) selection as in [Otero-Millan et al. 2014] to ensure balanced clusters.

3.1 Pre-processing

Pre-processing is critical in our dataset because of the challenges of bedside recordings. Our strategy is to first eliminate any portion of the eye movement signal that does not correspond with true eye movements and instead may come from erroneous tracking of the pupil due to full or partial pupil occlusion. This is done because the properties of noise and artifacts depend on the hardware and algorithms used for eye tracking; while the properties of eye movements should not depend to a great extent on the system and instead can be characterized for a given population of subjects or patients.

During pre-processing we consider both the horizontal (X) and vertical (Y) together and always remove noisy periods in both components at the same time. First, missing samples of periods less than 100ms are imputed using Steffen's method [Steffen 1990]. Next, periods of eye position, velocity, acceleration or jerk that are out of normal physiological ranges are removed (90°, $10^{3\circ}/s$, $5x10^{4\circ}/s^2$ and $5x10^{6\circ}/s^3$ respectively) with an additional 200ms before and after. These periods could be due to various reasons such as blinking, movement of goggles/ patient, problems in lightning, errors on pupil tracking, etc. Further, in some cases the data does not exceed those physiological ranges but instead presents a high frequency fluctuation which may be due to the pupil tracking being uncertain on finding the center of the pupil. These periods of fluctuating noise are detected using the fact that human eye cannot make more than 15 saccades in a second [Lee et al. 2002; Møller et al. 2002]. The eye movement signals are windowed into periods of 200ms and periods with a number of peaks of velocity (greater than $20^\circ/s$) that exceed this physiological limit are detected and removed accordingly. The pre-processed signal is then resampled to 500Hz to improve the precision of the detection of the beginning and end of the saccades/ quick-phases.

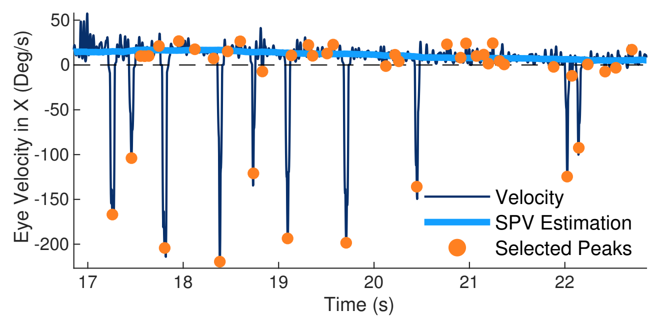

Figure 2: Initial SPV estimation and subtraction from eye velocity.

3.2 Peak Detection and Selection

In the presence of nystagmus, the velocity of the eye may never be close to zero. During quick-phases, the eye will reach velocities typical of saccades and during slow-phases the eye will move in the opposite directions with a slow-phase velocity (SPV) that can go up to $100^\circ/s$ and thus overlapping with typical velocities of small saccades [Otero-Millan et al. 2008]. To identify the quick-phases, we will first obtain a rough estimate of the SPV and subtract it from our signal (Figure 2). This will ensure that the velocity peaks found during slow phase have a distribution centered around zero while the quick-phase peaks remain far from zero and thus improve the result of clustering. To estimate the SPV, we iteratively exclude the highest 5% (1% at a time) velocity samples in a running window of 1s and then apply a median filter to the remaining samples. This initial low pass velocity (estimate of baseline SPV) is then subtracted from the instantaneous eye velocity. Physiologically, humans have a minimum intersaccadic interval of around 30ms [Otero-Millan et al. 2008]. Thus, we select the velocity peaks (from each of the two components) that are separated by at least 30ms.

In order to provide an approximately equal number of peaks that correspond with quick phases and peaks that correspond with noise to the clustering algorithm, we first obtain the rate R_1 of what could be considered clear saccade peaks. We defined those as the peaks with prominence above λ times the median absolute deviation of the prominence of all the peaks, with a similar approach as the method from Engbert and Kliegl to detect saccades [Engbert and Kliegl 2003]. We used a value of λ equal to 5. We then define the R_2 rate as $r \times R_1$ with r=3. Then, the set of peaks we will feed to the cluster algorithm will be the fastest peaks in the data so as to obtain a rate of R_2 peaks per second. This set should include all or most of the QPs and a comparable number of noise peaks. As stated previously, the rate factor is based on the physiological observation that saccades rates are limited to 1-6 per second during most tasks. Thus, the peak selection ensures that the expected clusters are balanced, aiding in the subsequent clustering stage.

3.3 Clustering

Three different features are selected and computed from the selected peaks. These are:

- Prominence computed as maximum height of the peak from the baseline
- Difference between peak velocity and median of eye velocity 100ms before/ after the peak that signifies the noise level around the velocity peak
- 1^{st} level decomposition coefficients of discrete wavelet transform (using 10^{th} order Daubechies wavelet, due to its similarity with the Gaussian shape) of the velocity peaks

The extracted features are then normalized and clustered using the Spectral clustering approach [Ng et al. 2002]. This type of clustering is less affected by the shape of the clusters than traditional clustering approaches such as k-means. In this approach, the similarity between data points is represented using a graph Laplacian matrix [von Luxburg 2007], $G = (F, E)$, where F is the set of feature vectors (data points) and E is the set of edges between them. The data points represent nodes of the graph and the strength of the edge connecting any two graph nodes, f_i and f_j, is computed using a Gaussian similarity function, $s(f_i, f_j) = exp(\frac{-|f_i - f_j|^2}{2\sigma^2})$. This essentially depicts the similarity between the two nodes and σ controls for the width of the overlapping neighborhood between data points. For the present study, any value of σ greater than 3 did not produce differences in the results obtained. Eigen vectors of the normalized version of the graph Laplacian then represent the two clusters.

4 DATA RESOURCE

We evaluated our new method with a nystagmus dataset collected as part of the AVERT clinical trial (Acute Video-oculography for Vertigo in Emergency Rooms for Rapid Triage), which is an ongoing multicenter study, started in 2015, comparing the accuracy of eye-movement based diagnosis and standard diagnosis. Patients presenting with acute vertigo or dizziness were subjected to a battery of tests including the Dix-Hallpike test (vision denied; left and right). Data was collected using a portable VOG (Video-Oculography) device (Figure 3), ICS Impulse (GN Otometrics^{TM}, Taastrup, Denmark) with a frame rate of 60Hz. Eye position in the device was calibrated using laser targets projected from the goggles. This dataset consists of recordings from 103 patients, with a total of 157 recordings of the Dix-Hallpike tests. Each recording is at least 30s long and is monocular (obtained from only right eye). We evaluated our method on a subset of the nystagmus data consisting of 30 randomly sampled patient recordings with a total time of 1040s (about 30s per recording) containing 1734 labelled QPs. This selection of data was manually labelled for fixations and saccades/ QPs by one of the authors, SAP.

For further validation of our method, we used a saccade dataset without nystagmus [Andersson et al. 2017; Larsson et al. 2013]. This data (referred to as saccade dataset) was collected using the SensoMotoric Instruments Hi-Speed 1250 at 500Hz.

Figure 3: ICS Impulse VOG Goggles used in AVERT Trial for Data Collection.

5 RESULTS

We evaluated the new algorithm in the nystagmus and saccade datasets described above by comparing its performance with two published state-of-the-art eye detection methods.

First, a method using a one dimensional convolutional neural network (CNN) in combination with a bidirectional long short-term memory has shown to outperform 12 other eye movement detection algorithms [Startsev et al. 2019]. The CNN has been trained using over 4h of GazeCom recordings [Dorr et al. 2010] that have been collected at 250Hz. The best model obtained in this method was also used to test on the saccade dataset [Andersson et al. 2017] (collected at 500Hz), without the need for retraining. For the present study, the 1D CNN-BLSTM has been used directly without training and tested using speed and direction features, as per the recommendation in the study [Startsev et al. 2019], for both the nystagmus dataset as well as the saccade dataset.

Second, a method that has employed similar approach using neural networks, U'N'Eye [Bellet et al. 2018]. This CNN has been developed to be accurate with minimal amount of training data. The U'N'Eye has been found to outperform other algorithms with just 50s of training data. For the present study, the U'N'Eye webservice was used to train the network using 50s, 220s and 830s of the nystagmus dataset. Due to limited data (using the video stimulus) from the saccade dataset, it was not possible to evaluate the U'N'Eye on this data. The recordings from the nystagmus dataset used for training U'N'Eye are also randomly sampled and exclusive to those used for testing.

In our approach to detection there are only two possible types of eye movements: quick-phases and slow-phases. Generally, quick-phases include saccades and slow-phases include all smooth eye movements that occur between saccades (fixation, smooth pursuit, vestibulo-ocular reflex, slow-phases of spontaneous nystagmus).

For this reason, we combined the labels from pursuit and fixation into a single type of eye movement (fixation) in the saccade dataset and in the outputs from the two state-of-the-art methods.

Figure 4 shows an example of a recording with nystagmus and the results obtained with each method and the manual labels. Table 1 shows the event-level F_1 scores (harmonic mean of precision and recall) obtained for the 3 methods when tested on the nystagmus dataset. The U'N'Eye has been trained using the indicated amount of training data from the nystagmus dataset via the U'N'Eye online web service. Similarly, Table 2 shows the event-level F_1 scores obtained for the proposed QP detection method and the 1D BLSTM CNN using the saccade dataset. Since many of the parameters are derived from known physiology of the ocular motor system, no parameter tuning was done for the proposed method when evaluated on either of the datasets.

We found that our method (QP) performs better in our patient nystagmus dataset when compared with the other two methods. On the other hand, in the saccade dataset of healthy subjects, the neural network method performs only slightly better than our method.

Table 1: Comparison on nystagmus dataset.

Method	Event-level F1	
	Saccade	Fixation
QP	0.79	0.88
U'N'Eye CNN (830s)	0.67	0.98
U'N'Eye CNN (220s)	0.62	0.98
U'N'Eye CNN (50s)	0.50	0.97
1D CNN-BLSTM	0.35	0.91

Table 2: Comparison on saccade dataset.

Method	Event-level F1	
	Saccade	Fixation
1D CNN-BLSTM	0.88	0.93
QP	0.86	0.84

6 DISCUSSION

Present study proposes a novel approach to saccade/ QP detection intended to work in the presence of nystagmus and in the challenging bedside environments. Importantly, the proposed method is modular, allowing for different stages to be improved independently. Moreover, the use of unsupervised clustering eliminates laborious process of manual labeling and model training or optimizing parameters.

Results of evaluating proposed algorithm with two other state-of-the-art methods on the nystagmus dataset are shown in Table 1. Although present method is shown to outperform the neural networks in saccade detection on the nystagmus patient data, the F_1 scores are only favorable. This can be attributed to the challenges of the nystagmus data that is recorded from bedside in hospitals, in contrast to controlled laboratory conditions. Table 2 shows the results when the proposed method is applied on other data, without any parameter tuning. The results are found to be comparable to that of the state-of-the-art eye detection methods.

Event detection in data from patients presents many challenges. First, the physiological properties of eye movements may change due to the conditions of the patient. That may make statistical, or neural network models built with data from any other population not valid. Second, patients may not be ideal subjects, with poor cooperation, and indirect effects of their condition affecting eyelid closure, attention, or steadiness of posture. Finally, many clinics do not operate expensive eye tracking devices that can record eye movement data at high sampling rate, instead low sampling rate such as 60Hz in our dataset may be common. For all these reasons, we propose a method that can be robust to all these challenges and work well even if not optimally in most datasets.

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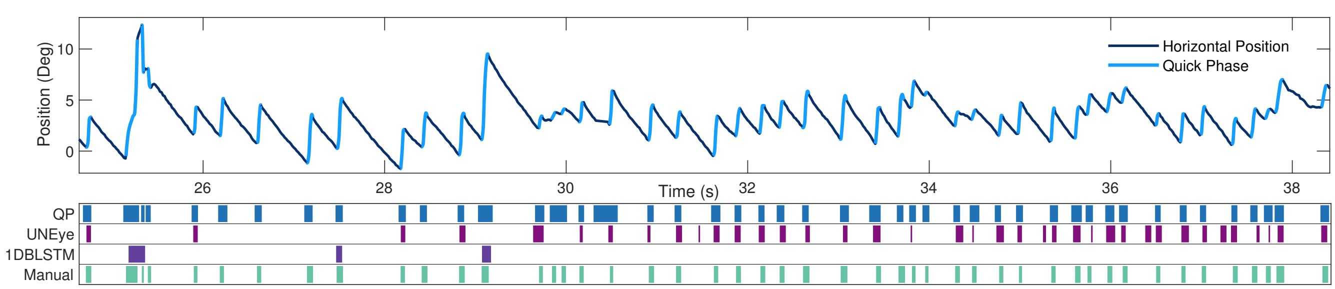

Figure 4: Results on a recording from the nystagmus dataset using proposed Quick-Phase Detection method (QP), 1D BLSTM CNN and the U'N'EYE CNN with indicated manual labels of the quick-phases

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